SENTINELQ

AN INTELLIGENT HARDWARE GUARDIAN FOR IoT SECURITY

EDGE AI | PHYSICAL SENSING | AUTONOMOUS RESPONSE | IoT SECURITY

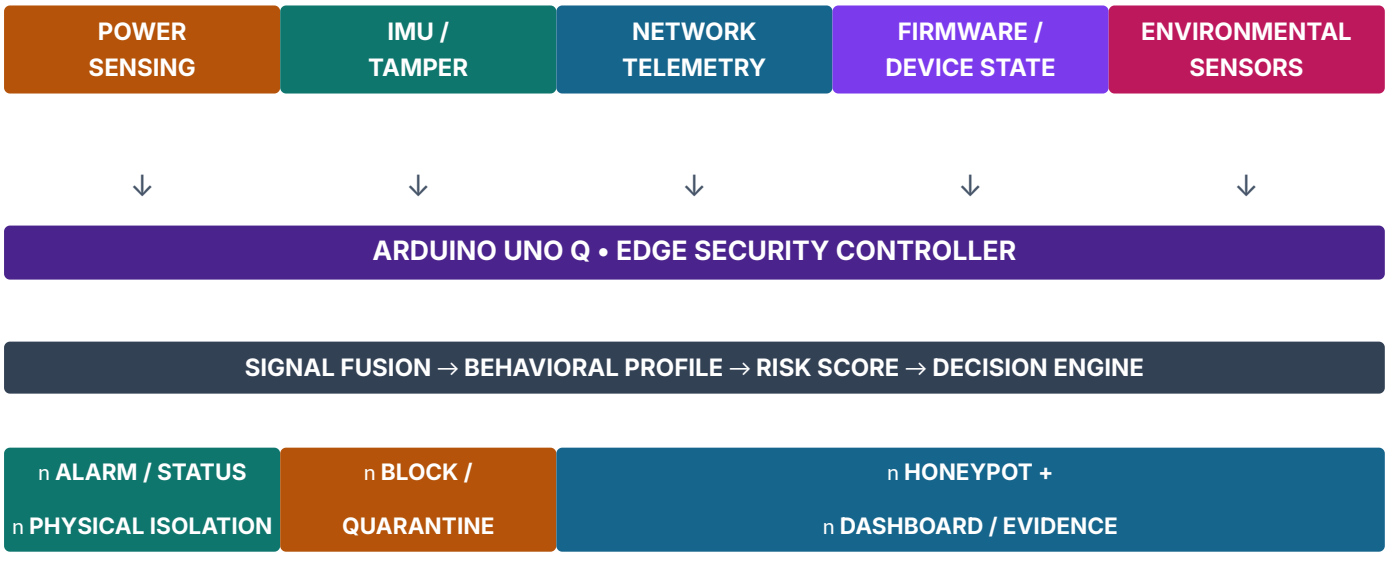
1 PROJECT CONCEPT

Modern IoT devices can be compromised through network attacks, physical tampering, abnormal power consumption, firmware manipulation, and unauthorized behavioral changes. SentinelQ is a hardware-centric cyber-physical security system that uses the Arduino Uno Q as an intelligent edge security controller to detect, analyze, and autonomously respond to such threats.

2 METHODOLOGY

SentinelQ continuously monitors a protected IoT node using multiple physical and digital signals. Current/power sensing captures electrical behavior, an IMU detects vibration or physical tampering, device-state and firmware checks identify integrity violations, and network telemetry captures communication anomalies. The Uno Q fuses these heterogeneous signals into a behavioral profile and computes a dynamic threat score using embedded rules combined with machine-learning-based anomaly detection. Multi-signal correlation reduces false alarms and identifies coordinated compromise. When the threat score crosses predefined levels, SentinelQ can trigger visual/audible alarms, rate-limit or block suspicious communication, and physically isolate the affected device through a switching interface. In a controlled laboratory environment, suspicious sessions can additionally be redirected to a honeypot for evidence collection and behavioral analysis. A companion dashboard presents device state, anomalies, risk score, attack timeline, and response actions.

3 PROPOSED HARDWARE BLOCK DIAGRAM



4 PROPOSED OUTCOME

The project aims to demonstrate that hardware can serve as an active security layer rather than merely a sensing platform. SentinelQ will provide real-time detection, explainable threat assessment, and physical containment of compromised IoT devices. Its multimodal approach is expected to improve detection confidence compared with network-only monitoring while providing a tangible, live hardware response suitable for resource-constrained IoT environments.

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